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Flexible HV Connector Busbar Interconnect System

Abstract

A busbar interconnect system having a housing of an inverter, and an interconnect assembly mounted in the housing, where the interconnect assembly includes an insulator housing, a first busbar having first and second ends, a second busbar having first and second ends, and the first busbar, the second busbar, and a mounting bracket at least partially surrounded by the insulator housing, the mounting bracket mounted to the housing. In a first configuration, a first of a plurality of connectors is connected to the first end of the first busbar and a second of the plurality of connectors is connected to the first end of the second busbar. In a second configuration, the first of the plurality of connectors is connected to the second end of the first busbar and the second of the plurality of connectors is connected to the second end of the second busbar.

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Background/Summary

FIELD OF THE INVENTION

[0001] The invention relates generally to a busbar interconnect system for an inverter, which facilitates the connection between a high-voltage DC battery and a high-voltage inverter.

BACKGROUND OF THE INVENTION

[0002] Power inverters for hybrid electric and battery electric vehicles are generally known. Current high-voltage (HV) inverters have non-configurable battery input/output. Therefore, there is only one possible connection between the DC connectors of battery cables and the inverter. This limitation of how the DC connectors may be connected to the inverter limits the various applications an HV inverter may be used for. Hybrid electric and battery electric vehicles may have different design restraints and packaging requirements.

[0003] Accordingly, there exists a need for configurable connection between an HV electronic box and a battery, which is suitable for multiple applications, provides for multiple orientations and connections between the HV box and the battery, and also meets various packaging requirements.

SUMMARY OF THE INVENTION

[0004] In an embodiment, the present invention is a busbar interconnect system for an inverter. In an embodiment, the busbar interconnect system provides one or more possible connections between a high-voltage (HV) DC battery and HV inverter in a vehicle.

[0005] The busbar interconnect system of the present invention improves transmitting power efficiency, reduces weight, volume, cost and complexity of power management systems, as well as simplifies and improves flexibility of the assembly process. The busbar interconnect system is suitable for use with a HV inverter and vehicles having various powertrain architectures.

[0006] In an embodiment, the busbar interconnect system of the present invention includes an interconnect assembly having two busbars, a plastic holder (such as an insulator housing), several fasteners (such as a self-clinching or weld fastener), and a mounting feature to fasten the interconnect assembly to the inverter housing.

[0007] In an embodiment, the terminals of the DC connector may be connected to the interconnect assembly in one of multiple ways, such that it is only necessary to change the location of where the terminals are connected to the busbars to accommodate the different vehicle applications.

[0008] The busbar interconnect system of the present invention facilitates ease of installation by using standard screw sizes, providing functionality that minimizes complexity during the assembly process.

[0009] The busbar interconnect system of the present invention includes the use of busbars which are shaped to reserve sufficient design margin to mount an HV battery connector, such that the bearing surface of the busbars in contact with the HV battery connector is suitable to facilitate the current load of the HV battery connector.

[0010] The busbar interconnect system of the present invention allows for the flexibility of connecting the HV DC input to the inverter in multiple locations.

[0011] In an embodiment, the busbar interconnect system of the present invention

[0012] In one or more embodiments, the busbar interconnect system of the present invention may be used for applications such as HV DC or AC connection points.

[0013] The busbar interconnect system of the present invention provides a versatile (i.e., flexible connection) connection solution between an HV DC battery and an HV inverter. The busbar interconnect system of the present invention includes the fastening of HV DC connectors to the inverter in various positions which allows for accommodation of various packaging requirements of the vehicle.

[0014] In an embodiment, the busbar interconnect system of the present invention includes

connection points located on opposite sides of inverter.

[0015] In an embodiment, the present invention is a busbar interconnect system, having a housing of an inverter, and an interconnect assembly mounted in the housing, where the interconnect assembly includes an insulator housing, a first busbar having a first end and a second end, the first busbar at least partially surrounded by the insulator housing, a second busbar having a first end and a second end, the second busbar at least partially surrounded by the insulator housing, and a mounting bracket at least partially surrounded by the insulator housing, the mounting bracket mounted to the housing.

[0016] In a first configuration, a first of a plurality of connectors is connected to the first end of the first busbar and a second of the plurality of connectors is connected to the first end of the second busbar. In a second configuration, the first of the plurality of connectors is connected to the second end of the first busbar and the second of the plurality of connectors is connected to the second end of the second busbar.

[0017] In an embodiment, the housing of the inverter includes a first housing portion, a second housing portion connected to the first housing portion and forming a cavity, the interconnect assembly disposed in the cavity. In an embodiment, a first mounting boss is integrally formed as part of the second housing portion and a second mounting boss is integrally formed as part of the second housing portion such that the mounting bracket is connected to the first mounting boss and the second mounting boss.

[0018] In an embodiment, the mounting bracket includes a first leg, which protrudes from the insulator housing, a first mounting flange integrally formed with the first leg, a second leg, which protrudes from the insulator housing, and a second mounting flange integrally formed with the second leg. In an embodiment, the first mounting flange is connected to the first mounting boss, and the second mounting flange is connected to the second mounting boss.

[0019] In an embodiment, the first housing portion of the inverter includes a first pair of apertures integrally formed as part of the first housing portion, and a second pair of apertures integrally formed as part of the first housing portion.

[0020] In the first configuration, the first of the plurality of connectors extends through one of the first pair of apertures when the first of the plurality of connectors is connected to the first end of the first busbar, and the second of the plurality of connectors extends through another of the first pair of apertures when the second of the plurality of connectors is connected to the first end of the second busbar.

[0021] In the second configuration, the first of the plurality of connectors extends through one of the second pair of apertures when the first of the plurality of connectors is connected to the second end of the first busbar, and the second of the plurality of connectors extends through another of the second pair of apertures when the second of the plurality of connectors is connected to the second end of the second busbar.

[0022] In an embodiment, each of the plurality of connectors includes a terminal having an assembly aperture, and the terminal is connected to one of the first busbar or the second busbar when the plurality of connectors are in the first configuration or the second configuration.

[0023] Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The present invention will become more fully understood from the detailed description and

the accompanying drawings, wherein:

[0025] FIG. 1 is a perspective view of a high voltage inverter having a busbar interconnect system, according to embodiments of the present invention;

[0026] FIG. 2 is a perspective view of a high voltage inverter having a busbar interconnect system, with the cover of the inverter removed, according to embodiments of the present invention;

[0027] FIG. 3A is a perspective view of an interconnect assembly which is part of a busbar interconnect system, according to embodiments of the present invention;

[0028] FIG. 3B is a top view of an interconnect assembly which is part of a busbar interconnect system, according to embodiments of the present invention;

[0029] FIG. 3C is a side view of an interconnect assembly which is part of a busbar interconnect system, according to embodiments of the present invention;

[0030] FIG. 4A is a sectional view taken along lines 4A-4A of FIG. 2, prior to the fasteners being inserted into mounting bosses of the second housing portion of the inverter, and prior to the first housing portion being connected to the second housing portion;

[0031] FIG. 4B is a sectional view taken along lines 4A-4A of FIG. 2, after the fasteners have been inserted into mounting bosses of the second housing portion of the inverter, and prior to the first housing portion of the inverter being connected to the second housing portion of the inverter;

[0032] FIG. 4C is a sectional view taken along lines 4A-4A of FIG. 2, after the fasteners have been inserted into mounting bosses of the second housing portion of the inverter, and after the first housing portion of the inverter is connected to the second housing portion of the inverter;

[0033] FIG. 4D is a sectional view taken along lines 4D-4D of FIG. 2;

[0034] FIG. 5A is a top view of a high voltage inverter having a busbar interconnect system, with the cover of the inverter removed and a pair of DC connectors connected to the interconnect assembly in a first configuration, according to embodiments of the present invention; and

[0035] FIG. 5B is a top view of a high voltage inverter having a busbar interconnect system, with the cover of the inverter removed and a pair of DC connectors connected to the interconnect assembly in a second configuration, according to embodiments of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0036] The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

[0037] An inverter having a busbar interconnect system according to the present invention is shown in FIG. 1, generally at **10**. Referring to FIGS. 1-2, the inverter includes a housing, shown generally at **12**, having first housing portion **14a** and a second housing portion **14b** and disposed in a cavity, shown generally at **16**, of the housing **12** is an electromagnetic interference (EMI) filter assembly, shown generally at **18**.

[0038] Also disposed in the cavity **16** and in electrical communication with the EMI filter assembly **18** is a first busbar **20** and a second busbar **22**. The busbars **20,22** are part of an interconnect assembly, shown generally at **24**. The first housing portion **14a** includes an aperture, shown generally at **26**, and the interconnect assembly **24** is located in the cavity **16** such that the interconnect assembly **24** may be seen when looking into the aperture **26**, as shown in FIG. 2.

[0039] Referring to FIGS. 3A-3C, the interconnect assembly **24** includes an insulator housing **28**, which functions as an insulator. The busbars **20,22** extend through the insulator housing **28**, and the busbars **20,22** are spaced apart from one another to provide a sufficient clearance, or air gap, to prevent undesirable arcing. In an embodiment, the air gap between the busbars **20,22** may be 8 mm, but it is within the scope of the invention that the air gap may be more or less to accommodate different voltages, allowable pollution levels, and various required creepages/clearances, depending upon the application.

[0040] The busbars **20,22** also include various mounting apertures **30** and connected to the busbars **20,22** are connection features **32**, and each of the connection feature **32** is press-fit into a corresponding mounting aperture **30**, shown in FIG. 3A-3C. More specifically, in an embodiment,

each of the connection features **32** is generally cylindrical in shape and includes a corresponding flange portion (not shown), which is pressed into a corresponding mounting aperture **30**, and each of the connection features **32** also has an internal threaded surface, which is used for connection with a fastener, such as a screw or bolt. In other embodiments the connection features **32** may be a self-clinching fastener or weld fastener.

[0041] The interconnect assembly **24** also includes a mounting bracket **34**. The mounting bracket **34** includes a first leg **36a** and a second leg **36b**, and extending from the first leg **36a** is a first mounting flange **38a**, and extending from the second leg **36b** is a second mounting flange **38b**. There is also a mounting aperture **40a,40b** integrally formed as part of each mounting flange **38a,38b**, respectively.

[0042] Referring to FIGS. **4A-4D**, during assembly, the interconnect assembly **24** is attached to two mounting bosses **42,44** located in the second housing portion **14b**. More specifically, there is a first fastener **46a** and a second fastener **46b**, which in this embodiment are bolts, where the first fastener **46a** is inserted through the mounting aperture **40a** and into a corresponding threaded aperture of the first mounting boss **42**, and the second fastener **46a** is inserted through the mounting aperture **40b** and into a corresponding threaded aperture of the second mounting boss **44**. The fasteners **46a,46b** inserted into the apertures of the mounting bosses **42,44** is shown in FIG. **4B**.

[0043] Once the interconnect assembly **24** is mounted as shown in FIG. **4B**, the first housing portion **14a** is then connected to the second housing portion **14b**. Referring to FIG. **4C**, more specifically, another fastener **48** is inserted through an aperture **50** integrally formed as part of a flange portion **52**, the fastener **48** is then inserted into a threaded aperture of a mounting protrusion **54**. The flange portion **52** is part of the first housing portion **14a**, and the mounting protrusion **54** is part of the second housing portion **14b**. Referring to FIGS. **1-2**, there are additional fasteners used in a similar manner to connect the first housing portion **14a** to the second housing portion **14b**.

[0044] Once the first housing portion **14a** is connected to the second housing portion **14b**, a pair of DC connectors **56a,56b** may then be connected to the interconnect assembly **24**. Referring to FIGS. **1-2, 4C-4D**, and **5A-5B**, the first housing portion **14a** includes a first pair of connector apertures **58a,58b** and a second pair of connector apertures **60a,60b**. In a first configuration, shown in FIG. **2**, the first DC connector **56a** is connected the first housing portion **14a** such that the first DC connector **56a** extends through the first connector aperture **58a** and is connected to a first end **62a** the first bus bar **20**. Also, in the first configuration, the second DC connector **56b** is connected to the first housing portion **14a** such that the second DC connector **56b** extends through the second connector aperture **58b** and is connected to a first end **64a** the second bus bar **22**. More specifically, each DC connector **56a,56b** includes a corresponding terminal, and an assembly aperture is formed as part of the terminal. In a non-limiting example, a fastener **70a** is inserted through the assembly aperture **66** of the terminal **68** of the DC connector **56a**, through the mounting aperture **30**, and into the connection feature **32** connected to the first end **62a** of the busbar **20**, as shown in FIGS. **2, 4D** and **5A**. The terminal of the second DC connector **56b** is connected to the first end **64a** of the second busbar **22** in a similar manner.

[0045] In a second configuration, the DC connectors **56a,56b** may be connected to the second ends **62b,64b** of the busbars **20,22**, respectively. More specifically, in the second configuration, shown in FIG. **5B**, first DC connector **56a** is connected the first housing portion **14a** such that the first DC connector **56a** extends through the first connector aperture **60a** and is connected to the second end **62b** the first bus bar **20**. Also, in the second configuration, the second DC connector **56b** is connected to the first housing portion **14a** such that the second DC connector **56b** extends through the second connector aperture **60b** and is connected to a second end **64b** the second bus bar **22**. In a similar manner as described above, the fastener **70a** is inserted through the assembly aperture **66** of the terminal **68** of the DC connector **56a**, through the mounting aperture **30**, and into the connection feature **32** connected to the second end **62b** of the busbar **20**, as shown in FIG. **5B**. The terminal of the second DC connector **56b** is connected to the second end **64b** of the second busbar

22 in a similar manner using the fastener **70b**.

[0046] Mounting the DC connectors **56a,56b** in one of the two configurations described above provides a connection between the high-voltage (HV) DC battery the HV inverter **10** in a vehicle. The interconnect assembly **24** is suitable for use with a common HV inverter and vehicles having various powertrain architectures.

[0047] The use of the interconnect assembly **24** facilitates only necessitating the change of the location of the DC connectors **56a,56b** to accommodate different vehicle applications.

[0048] The use of the interconnect assembly **24** also allows for the flexibility of installing the HV DC input into the inverter **10** in multiple locations.

[0049] The description of the invention is merely exemplary in nature and, thus, variations that do not depart from the gist of the invention are intended to be within the scope of the invention. Such variations are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. An apparatus, comprising: a busbar interconnect system, comprising: a housing of an inverter; an interconnect assembly mounting in the housing, the interconnect assembly further comprising: an insulator housing; a first busbar having a first end and a second end, the first busbar at least partially surrounded by the insulator housing; a second busbar having a first end and a second end, the second busbar at least partially surrounded by the insulator housing; a mounting bracket at least partially surrounded by the insulator housing, the mounting bracket mounted to the housing; wherein, in a first configuration, a first of a plurality of connectors is connected to the first end of the first busbar and a second of the plurality of connectors is connected to the first end of the second busbar, and in a second configuration, the first of the plurality of connectors is connected to the second end of the first busbar and the second of the plurality of connectors is connected to the second end of the second busbar.

2. The apparatus of claim 1, the housing of the inverter further comprising: a first housing portion; a second housing portion connected to the first housing portion and forming a cavity, the interconnect assembly disposed in the cavity; a first mounting boss integrally formed as part of the second housing portion; and a second mounting boss integrally formed as part of the second housing portion; wherein the mounting bracket is connected to the first mounting boss and the second mounting boss.

3. The apparatus of claim 2, the mounting bracket further comprising: a first leg, which protrudes from the insulator housing; a first mounting flange integrally formed with the first leg; a second leg, which protrudes from the insulator housing; a second mounting flange integrally formed with the second leg; wherein the first mounting flange is connected to the first mounting boss, and the second mounting flange is connected to the second mounting boss.

4. The apparatus of claim 2, the first housing portion of the inverter further comprising: a first pair of apertures integrally formed as part of the first housing portion; and a second pair of apertures integrally formed as part of the first housing portion; wherein, in the first configuration, the first of the plurality of connectors extends through one of the first pair of apertures when the first of the plurality of connectors is connected to the first end of the first busbar, and the second of the plurality of connectors extends through another of the first pair of apertures when the second of the plurality of connectors is connected to the first end of the second busbar, wherein, in the second configuration, the first of the plurality of connectors extends through one of the second pair of apertures when the first of the plurality of connectors is connected to the second end of the first busbar, and the second of the plurality of connectors extends through another of the second pair of apertures when the second of the plurality of connectors is connected to the second end of the second busbar.

5. The apparatus of claim 1, each of the plurality of connectors further comprising: a terminal

having an assembly aperture; wherein the terminal is connected to one of the first busbar or the second busbar when the plurality of connectors are in the first configuration or the second configuration.

6. An interconnect assembly for an inverter, comprising: an insulator housing; a first busbar having a first end and a second end, the first end of the first busbar protruding out of the opposite side of the insulator housing as the second end of the first busbar; a second busbar having a first end and a second end, the first end of the second busbar protruding out of the opposite side of the insulator housing as the second end of the second busbar; and a mounting bracket extending through the insulator housing, the mounting bracket connected to a housing of an inverter; wherein, in a first configuration, a first of a plurality of connectors is connected to the first end of the first busbar and a second of the plurality of connectors is connected to the first end of the second busbar, and in a second configuration, the first of the plurality of connectors is connected to the second end of the first busbar and the second of the plurality of connectors is connected to the second end of the second busbar.

7. The interconnect assembly for an inverter of claim 6, the mounting bracket further comprising: a first leg; a first mounting flange integrally formed with the first leg; a second leg; and a second mounting flange integrally formed with the second leg; wherein the first leg and the second leg protrude from the insulator housing.

8. The interconnect assembly for an inverter of claim 7, the housing of the inverter further comprising: a first housing portion; a second housing portion connected to the first housing portion and forming a cavity, the interconnect assembly disposed in the cavity; a first mounting boss integrally formed as part of the second housing portion; and a second mounting boss integrally formed as part of the second housing portion; wherein the first mounting flange is connected to the first mounting boss, and the second mounting flange is connected to the second mounting boss.

9. The interconnect assembly for an inverter of claim 8, the first housing portion of the inverter further comprising: a first pair of apertures integrally formed as part of the first housing portion; and a second pair of apertures integrally formed as part of the first housing portion; wherein, in the first configuration, the first of the plurality of connectors extends through one of the first pair of apertures when the first of the plurality of connectors is connected to the first end of the first busbar, and the second of the plurality of connectors extends through another of the first pair of apertures when the second of the plurality of connectors is connected to the first end of the second busbar, wherein, in the second configuration, the first of the plurality of connectors extends through one of the second pair of apertures when the first of the plurality of connectors is connected to the second end of the first busbar, and the second of the plurality of connectors extends through another of the second pair of apertures when the second of the plurality of connectors is connected to the second end of the second busbar.

10. The interconnect assembly for an inverter of claim 6, each of the plurality of connectors further comprising: a terminal having an assembly aperture; wherein the terminal is connected to one of the first busbar or the second busbar when the plurality of connectors are in the first configuration or the second configuration.
